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Creativity, Expectancy Violations, and Impression Formation: Effects of Novelty and Appropriateness in Online Dating Profile Texts

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ABSTRACT

This experimental study investigates how novelty and appropriateness – the two criteria of creativity – affect impressions formed of online dating profile owners. More specifically, it examines how both criteria influence perceived romantic attractiveness, and whether expectancy violations and person perceptions (i.e., originality and oddness) mediate these effects. Participants ($N = 295$) were members of a dating platform who were presented with dating profile texts that met one or both criteria: (a) not novel and appropriate, (b) novel and appropriate, or (c) novel and inappropriate. In line with the expectations, results show that novel texts are less expected, and that their writers are perceived as more original and, in turn, as more romantically attractive. Writers of novel and non-novel texts are evaluated as equally attractive though, suggesting that appropriateness is a stronger determinant of romantic attractiveness. That is, novel texts that are inappropriate violate expectations more negatively than those that are appropriate, increasing oddness and, accordingly, decreasing romantic attractiveness. Findings suggest that novelty and appropriateness are two important creativity dimensions that explain effects of dating profile texts on impression formation. Moreover, we show that the expectancy violations theory is a useful framework to study creativity and impression formation in online dating.

The growing popularity of online dating also means an increasingly competitive environment to get your dating profile noticed and liked (Markowitz et al., 2018). People spend little time scrutinizing a dating profile and make the decision to like or dislike a potential partner swiftly (D'Angelo & Toma, 2017). Although the profile picture is often the first and most important cue to evaluate a potential partner, research has shown that the profile text also plays an important role in impression formation and dating profile evaluation

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(Van der Zanden et al., 2022). Most dating profile texts, however, are fairly similar in content, listing mostly basic interests, and often fall into clichés (e.g., Masden & Edwards, 2015; Tong & Van der Heide, 2022, p. 69; Whitty, 2008).

One possible way to appear attractive is crafting a creative profile text. Van der Zanden et al. (2022) found that perceived profile text creativity related to perceived profile owner attractiveness. These results may be explained from an expectancy violations framework (EVT; Burgoon, 1993). A creative profile text is usually unexpected in online dating as most profile texts lack novelty and uniqueness. A creative profile text does not conform to this expected standard and may thus violate online daters' expectations, which in turn may affect their impressions of the profile owner.

Following EVT (Burgoon, 1993), this violation of expectancies could both positively and negatively influence these interpersonal impressions, dependent on whether the communicative outcome (here: profile text) is better (positive expectancy violation) or worse than expected (negative expectancy violation). Positive and negative expectancy violations can color impressions about the personality of the person responsible for the violation (e.g., D'Angelo & Van Der Heide, 2016; DelGreco & Denes, 2020; Ramirez & Wang, 2008). If the consensus among online daters is a common, generic profile, writers who deviate from this, and thus go against the expected, may be evaluated as original. On the other hand, violating expectancies may also result in the profile owner coming across as odd, particularly if the expectancy is negatively violated. Both perceptions of profile owner originality and oddness are in turn likely to influence perceived attractiveness (Van der Zanden et al., 2022).

Whether a profile owner is seen as original or odd thus likely depends on the creativity of the profile text. Stein (1953) was one of the first who formulated a broad definition of creativity: "the creative work is a novel work that is accepted as tenable or useful or satisfying by a group at some point in time" (p. 311). Still, scholars widely agree that novelty and appropriateness are the two criteria of creativity (Runco & Jaeger, 2012). Novelty (also referred to as originality, unusualness) is the first and most important criterion and refers to an idea or product being unusual and different from what is regularly encountered in that specific context. The second criterion is appropriateness (also referred to as usefulness, valuable): a novel idea or product should also be meaningful within the context for which it is produced. To illustrate, a job application letter that is fully written in rhyme is highly novel as such application letters appear infrequently but is at the same time inappropriate. However, if all applicants start to write rhyming application letters, it only takes time until recruiters consider it common, while it may still be novel in the online dating context where users rarely encounter rhyming profile texts.

Thus, creativity in online dating texts may differentially affect perceived romantic attractiveness through perceived originality and oddness depending on whether the profile text is both novel and appropriate or novel and

inappropriate. Novel profile texts violate expectancies because they violate the norm of what is common in this type of texts. However, whether this violation is deemed positive or negative depends on the valence of that violation (Burgoon, 1993). Novelty may be evaluated positively when it happens within the boundaries of appropriateness, resulting in higher perceived originality and higher romantic attraction, but negatively when it exceeds appropriateness, resulting in higher perceived oddness and lower romantic attraction.

Taken together, this paper has two main goals. The first goal is to understand the underlying processes that explain the relationship between creativity on the one hand and romantic attractiveness on the other hand by using the framework of expectancy violations that has been employed little in the context of online dating (profiles). The second goal is to investigate how novelty and appropriateness, the two dimensions of creativity, affect profile owner originality and oddness and, in turn, romantic attractiveness.

This study may benefit media psychologists and communication scholars for two reasons. First, there is hardly any research that investigates whether these two creativity dimensions affect interpersonal impression formation. If creativity is an effective communicative strategy in online dating, it may be important in explaining the effects of other communicative messages as well, such as social media posts or job application letters. Second, this study uses EVT to explain potential positive and negative effects of message characteristics, showing that EVT may be used to study not only interpersonal interaction, but also the wider effect of communicative messages.

In this study, members of a Dutch dating platform on which profile texts play a significant role, read and assess a dating profile text in one of the three profile text conditions: (a) not novel and appropriate, (b) novel and appropriate, or (c) novel and inappropriate, after which they answer questions about the text, expectancy violations, and their perceptions about the profile writer (e.g., originality, oddness, romantic attractiveness).

Profile Texts and Impression Formation

Prior research on social networking sites consistently highlighted the role of profile elements in shaping initial impressions (e.g., Back et al., 2010; Gosling et al., 2002). Profile texts are no exception to this; texts and cues within them have demonstrated to influence how their writers are perceived in contexts of Twitter (Qiu et al., 2012), Facebook profiles, personal websites (Marcus et al., 2006; Vazire & Gosling, 2004), as well on dating profiles (Tong et al., 2020; Van der Zanden et al., 2020). An important reason for this strong reliance on texts for impression formation is the amount of information that language can give away about writers. For example, language errors impact person

perceptions of intelligence, attentiveness, and warmth, as well as perceptions of attractiveness (Van der Zanden et al., 2020).

Creativity in Online Dating Profiles

Creativity is another characteristic of a profile text that can affect impression formation but that yet garnered limited research focus. An exception is the study of Van der Zanden et al. (2022) who found that writers of profile texts that are rated as more original are evaluated more positively. More specifically, they found that writers of profile texts that are deemed more original are also evaluated as more intelligent, funnier, and, in turn, as more attractive and more interesting dating material. Moreover, a lack of (profile) text creativity or originality is one of the main criticisms users of dating platforms mention in interview studies, with profile texts often considered to be middle of the road and offering little in terms of in-depth and original content (Masden & Edwards, 2015; Whitty, 2008; Wobcke et al., 2015).

Creativity has been found to be a desired characteristic when assessing the attractiveness of a potential romantic partner (e.g., Doosje et al., 1999; Goodwin & Tinker, 2002; Rowatt et al., 2001; Watkins, 2017). One of the reasons may be that creative people are often considered to also possess more general cognitive abilities, such as understanding, remembering, and reasoning (Rosenberg & Tunney, 2008), making them more attractive romantic partners. In early stages of (potential) romantic relationship formation, original behavior was found to positively impact perceptions of interpersonal attraction (Apostolou & Christoforou, 2020; Gao et al., 2017; Watkins, 2017; White et al., 2018). For instance, Gao et al. (2017) found that men who complimented a woman's appearance by using metaphorical language (indicating novel language use) were evaluated as more creative and better dating material than men using literal language.

Novelty and Appropriateness as Dimensions of Creativity

Creativity is a multifaceted construct that is defined by two criteria: novelty and appropriateness (Runco & Jaeger, 2012). As there is no creativity without novelty, this dimension is seen as the first and most important criterion. It refers to something – in this study a dating profile text – being unique or uncommon. In general, novel ideas are invented by only a small selection of people and are therefore seen as distinct, uncommon, and/or unique (Runco & Charles, 1993).

While most studies highlight the importance of novelty in defining creativity, appropriateness plays a considerable role as well (Diedrich et al., 2015; White et al., 2018). Novelty is on its own not sufficient to describe creativity, as something that is only labeled as novel is not necessarily creative (Runco &

Jaeger, 2012). Creative ideas or products should also be appropriate and meaningful in order to be considered creative (e.g., Diedrich et al., 2015; Runco & Charles, 1993). Appropriateness refers to the usefulness or effectiveness of something, which means that a creative idea needs to fit the situation and needs to add value in the context for which it is produced (Runco & Jaeger, 2012). Creativity is thus a balancing act between novelty on the one hand and appropriateness on the other hand (e.g., Diedrich et al., 2015; Heinen & Johnson, 2018), but little is yet known about how the creativity dimensions influence impression formation, including perceptions of their producers.

Expectations in Online Dating

Drawing from the expectancy violations theory (Burgoon, 1993) and language expectancy theory (Burgoon et al., 2002), people's expectancies regarding communicative behavior and language use may affect evaluations and responses to this behavior, particularly when expectancies are violated. Burgoon and Hale (1988) argue that expectancies (hereafter used interchangeably with expectations) can be derived from multiple sources, such as the interaction partner, the relationship, and the context in which communication takes place. Within the context of online dating, these sources can for example be the profile text (e.g., interests listed in the profile text), expectancies based on (perceived) group identity derived from the profile picture (e.g., someone visiting a metal concert) or even the online dating platform itself (e.g., dating sites vs. hookup apps).

Only few studies have used EVT as a framework to understand impression formation in online dating, and those that did focused mainly on interpersonal communicative contexts. For example, Ramirez and Wang (2008) and Zhao and Yan (2022) investigated how discrepancies between an online dater's online and offline persona can affect impressions and relational outcomes. A study by DelGreco and Denes (2020) focused on gender-specific expectations during the initial interactional phase, concentrating on responses of women to compliments in message exchanges. In their study, participants were presented with a fictitious message exchange depicting a compliment to a woman ("Wow, you have gorgeous eyes!") and a response to this compliment that either affirmed expectations by just accepting the compliment ("Thank you!") or violated expectations. Their findings showed that women who violated expectations were evaluated as less likeable and socially attractive compared to those in the no violation condition.

The current study further tests the relevance of EVT by focusing on the profile evaluation phase prior to interaction. Based on encounters with prior profiles, online daters are likely to hold expectations about the content and form of other users' profile texts (Gibbs et al., 2006). They

are likely to expect profiles that are highly similar in terms of profile content and language use, for example by displaying the same general personality traits (e.g., “spontaneous”) and hobbies and interests (e.g., “love to travel”, “having fun with friends”; Masden & Edwards, 2015; Tong & Van der Heide, 2022; Whitty, 2008). Hence, profiles that deviate from this expected standard – that is, when they are novel – may not correspond with the expectancies of dating platform users. Thus, dating profiles written in line with the norms set in the online dating context will not violate expectations, while those that deviate from this norm will.

In an earlier study on novelty in online dating profile texts, Van der Zanden et al. (2022) conducted a content analysis on 308 online dating profile texts. In an earlier phase of the study participants scored the texts on perceived originality, and the authors linked these originality ratings to various coded text features. Two main predictors of perceived profile text originality were identified: metaphors and concrete self-disclosure. At this point in time, most dating profiles seemed to be written in literal language and include generic self-disclosures (“I like travelling”). Such dating profile texts are thus expected, while metaphors and concrete self-disclosure are relatively uncommon, hence unexpected. Thus, when a dating profile text includes metaphorical language and concrete personal information, this is a novel text that violates expectations more than a text that include literal language and generic personal information. This leads to the following hypothesis:¹

H1a. Novel online dating profile texts violate expectations more than profile texts that are not novel.

As corroborated by findings of DelGreco and Denes (2020) and Van der Zanden et al. (2022), communicative messages that are novel and thus deviate from expectancies can affect interpersonal impressions, for instance about the writer’s originality. However, the extent to which perceptions of profile owner originality explain potential relationships between text novelty, expectancy violations, and (romantic) attractiveness remains unexplored. We expect that expectancy violations explain the relationship between novel dating profile texts and perceptions of originality. In comparison to mainstream, middle-of-the-road profile texts, a novel text violates expectations of what is normally expected in profile texts. Thus, if a novel text is not perceived as distinct from the norm, this would not influence perceptions of the profile owner’s originality (positively). In sum, we expect that expectancy violations correlate positively with the perceived originality of the profile owner. Second, given that creative people – who thus display novel behaviors – are often regarded as more attractive (e.g., Doosje et al., 1999; Goodwin & Tinker, 2002; Rowatt et al., 2001; Watkins, 2017), these increased originality perceptions are likely to

subsequently enhance feelings of romantic attraction. This leads to the following two hypotheses: ²

H1b. Expectancy violations are positively related to perceptions of profile owner originality.

H1c. Perceived originality is positively related to perceived romantic attractiveness.

Positive and Negative Expectancy Violations in Dating Profile Texts

According to EVT, when there is a mismatch between expected and actual communicative behavior, expectancy value can be both positive and negative (Burgoon & Burgoon, 2001). While a positive expectancy violation occurs when actual behavior is better than the expected behavior, a negative expectancy violation occurs if the actual behavior is worse than the expected behavior.

While a novel dating profile text may violate expectations, the valence of this violation is likely to be determined by the appropriateness of the violation. Perceptions of what is (in)appropriate are generally based on social norms and conventions that emerge in specific (online) contexts (e.g., D'Angelo & Van der Heide, 2016, on picture self-presentation on various SNS; Lew & Walther, 2022, in a human and AI chatbot interaction). This suggests that these norms lead to expectations about what is appropriate and inappropriate behavior, also regarding users' dating profiles (Dredge & Anderson, 2021; Ellison et al., 2012; Kavroulaki, 2022). If expectations are violated because of an inappropriate profile text, this is likely to be a negative expectancy violation.

Even though there is not much known about online daters' specific normative expectancies of dating profiles, a positively framed and not overly intimate profile text is likely to be expected. That is because most online daters present a positive self, and they might expect others to do that as well (Ellison et al., 2012; Gibbs et al., 2006, 2011). Negative and/or too private and intimate self-disclosure may be particularly unexpected and perceived less appropriate in early stages of relationship formation (Caltabiano & Smithson, 1983; Chaikin & Derlega, 1974; Dredge & Anderson, 2021), as online daters are still strangers when browsing profiles of others. This suggests that concrete self-disclosures only improve creativity perceptions if this self-disclosed information is appropriate. In that case, concrete self-disclosure may be both novel and appropriate, leading to positive expectancy violations. However, if concrete self-disclosure is inappropriate by being negative and/or highly intimate, it may still be novel, but expectations are likely to be violated negatively.

Likewise, a metaphor may only be considered creative if it meets both the novelty and appropriateness criteria. If a metaphor is inappropriate, for example because the two metaphor domains badly link, leading to an incomprehensible metaphor (e.g., “Your face is a girl in hands of goddess;” Yang et al., 2019), the criterion of novelty may be met but not the one of appropriateness. Thus, while novelty (vs. not novel) presumably leads to higher expectancy violations scores in general, (in)appropriateness presumably determines if this violation is negative. We thus hypothesize:

H2a. Novel and inappropriate online dating texts violate expectations more negatively than novel and appropriate texts.

Negative expectancy violations – in our case as a result of a novel but inappropriate profile text – generally harm impressions formed of the person violating the expectations, for example regarding this person’s personality, social skills, and attractiveness (D’Angelo & Van Der Heide, 2016; DelGreco & Denes, 2020; Zhao & Yan, 2022). Owners of novel but inappropriate profiles may be perceived as more strange, odd, and/or peculiar, for instance because they do not seem to understand what is socially expected from them in the specific context. For example, White et al. (2018) found that flirting behavior that was novel in the way that it was different from what is commonly seen was rated more effective when it did not exceed appropriateness than when it did (e.g., telling potential romantic partners they look like a movie star vs. asking them to hold hands on a first date). Such inappropriate behavior can give a clue about this person’s general (lack of) feeling for social norms, and complicates predicting this person’s future behavior. It is therefore expected that if a text is novel but does not adhere to the criterion of appropriateness, violation valence will be negative. As a result, perceptions of romantic attractiveness will be lower because of increased oddness perceptions (Van der Zanden et al., 2022). In contrast, when the expectancy violation is positive (in case of novel, appropriate profile texts), this will reduce perceived oddness, thereby increasing romantic attractiveness. Therefore, we pose the following hypotheses:

H2b. Violation valence is negatively related to perceptions of profile owner oddness.

H2c. Perceived oddness is negatively related to perceived romantic attractiveness.

Method

Ethical clearance for this study was obtained in Spring 2022 by the Research Ethics and Data Management Committee of the Tilburg School of Humanities and Digital Sciences (REDC 2022.22). On OSF (<https://osf.io/sympn/>), the study's preregistration, materials, and dataset (including all measures) can be found.

Design

This online experimental study had a single-factor between-subjects design with three profile text conditions: (a) non-novel and appropriate, (b) novel and appropriate, and (c) novel and inappropriate. To investigate the effects of novelty and appropriateness separately, we compare not novel with novel dating profile texts (both appropriate and inappropriate), as well as novel texts that are inappropriate with those that are appropriate. As inappropriate texts will violate expectations of what can regularly be expected on a dating site, these texts will also always be (somewhat) novel. We did not include a not-novel, inappropriate condition as it would be impossible to operationalize texts that lack novelty but are at the same time inappropriate.

Separate mediators were considered relevant for both contrasts, that is, expectancy violations (M1) and profile owner originality (M2) for text novelty, and violation valence (M1) and profile owner oddness (M2) for text appropriateness. Perceived romantic attractiveness was the outcome variable for both models.

Participants

Participants were members of the Dutch dating site Parship NL, which presents itself as a dating site for finding a serious relationship. Parship NL helped with participant recruitment but was not involved in the study. Of all 365 participants who provided informed consent, 295 participants fully completed the online experiment. Of those 295 participants, 134 self-identified as a man (45.4%) and 161 (54.6%) as a woman. Participants were between 26 and 82 years old ($M_{age} = 55.4$, $SD_{age} = 11.7$), and except for two all of them were heterosexual. The majority (74.9%) had a college degree, the rest a vocational or high school degree (26.1%). Eleven participants (3.7%) indicated that Dutch was not one of their native languages.

Materials

To create the study's materials, we first created four different profile templates based on previous studies on online dating profile texts (Madsen & Edwards,

2015; Van der Zanden et al., 2018, 2022). Each template consisted of nine different topics: welcome, reason for online dating, two personality traits, two hobbies, type of relationship sought, type of partner sought, and a closing. Except for the welcome and closing, the topics appeared in a different order in each template. All topics in a template were different, but similar across the three text conditions of that template so that specific contents would not confound the effects. For example, in one template the two personality traits described were intelligent and caring at positions 3 and 6, while in another template the personality traits were extravert and social, at positions 5 and 6 respectively.

Once the templates were created, novelty and appropriateness were manipulated within them to create texts in the three conditions, resulting in a total of 12 profile texts (4 profile templates x 3 conditions) participants could be exposed to. Following Van der Zanden et al. (2022) who found that perceived creativity of online dating texts was mostly classified by (concrete) self-disclosure and metaphor use, we varied the use of these two elements to manipulate novelty and appropriateness. Specifically, all novel profile texts contained two metaphors and two concrete self-disclosure statements, which could either be appropriate or inappropriate depending on the condition.

Not novel and appropriate

The texts that were not novel and appropriate contained literal language, hence no metaphors, and contained only generic low intimate self-disclosure (e.g., “I like doing fun activities”, “I am looking for some with a good character”). As these generic self-disclosure statements are rather common on online dating platforms, these statements are not particularly unique and thus seen as appropriate. Table 1 explains in more detail how text novelty and appropriateness are manipulated across the three text conditions.

Novel and appropriate

All novel and appropriate texts contained two metaphors that were novel and appropriate. Conventional and cliché metaphors were avoided. Appropriate metaphors refer to those that are easily derivable from their context, so the reader could understand the analogy (Fraser, 1993; e.g., “I’m looking for someone as sweet as buttercup”). Moreover, all novel and appropriate texts contained two concrete self-disclosure statements, of which one was relatively low in intimacy (as opposed to a high intimate version in the inappropriate condition), and the other positive (as opposed to a negative version in the inappropriate condition).

Novel and inappropriate

The profile texts that were novel but inappropriate contained two metaphors that were novel but inappropriate at the same time. Inappropriate metaphors

Table 1. Definition, operationalization, and examples of the manipulations within the three text conditions.

	Not novel, appropriate	Novel, appropriate	Novel, inappropriate
Metaphors	Literal language <i>"I'm always friendly to others"</i>	Metaphor with clear analogy <i>"I'm as friendly as the big friendly giant of Roald Dahl"</i>	Metaphor with unclear analogy <i>"I'm as friendly as a blue Stabilo ballpoint"</i>
Concrete self-disclosure (SD)	Low intimate general SD <i>"If we go on a date, I would like to have fun while doing some nice activity together"</i>	Low intimate concrete SD <i>"If we go on a date, I would like to do this while enjoying a Verdejo and nachos from the oven"</i>	High intimate concrete SD <i>"If we go on a date, I would like to do this while enjoying a Verdejo and nachos from the oven, just like I did with my two previous dates"</i>
	Positive general SD <i>"I am adventurous and I regularly seek for unexpected and exciting situations"</i>	Positive concrete SD <i>"I am adventurous. In Guatemala I parachuted out of an airplane from 3142 meters."</i>	Negative concrete SD <i>I am adventurous. My hobby is skydiving and I'll not be quitting that to financially support you"</i>

are hard to understand within their context (Yang et al., 2019), which happens when there is no clear link between the metaphor's tenor and vehicle (Fraser, 1993). For example, "People regularly speak about me as an intelligent window of glass." Self-disclosure was manipulated by including a self-disclosure statement that was very high in intimacy and one that was negative, since both types of self-disclosures are perceived as inappropriate, particularly when people are still unacquainted (Caltabiano & Smithson, 1983; Chaikin & Derlega, 1974; Dredge & Anderson, 2021).

Pretest

Prior to the main experiment, a pretest among 66 participants (68.2% women; $M_{\text{age}} = 36$, $SD_{\text{age}} = 15.2$) was carried out to examine if the texts intended to be novel were indeed perceived as more novel, and the appropriate texts as more appropriate. Each pretest participant rated six online dating profile texts on their perceived text novelty and appropriateness on a scale from 1 (strongly disagree) to 7 (strongly agree).

Perceived text novelty was measured with three items (e.g., "This profile is unlike what others do;" see OSF for a list with all items; $\alpha = .87$). A one-way ANOVA with planned contrasts showed a main effect of text condition on perceived text novelty, $F(2, 187) = 55.95$, $p < .001$. Planned contrasts revealed that not novel (appropriate) texts ($M = 2.88$, $SD = 1.16$) scored significantly lower on text novelty than the other novel text conditions (novel, inappropriate: $M = 5.10$, $SD = 1.37$; novel, appropriate: $M = 4.83$, $SD = 1.34$), $t(187) = 10.52$, $p < .001$, $r = .61$). The two types of novel texts that differed in terms of appropriateness scored similarly on perceived text novelty, $t(187) = 1.15$, $p = .25$, $r = .08$.

Perceived text appropriateness was measured with another three items (e.g., "This profile text fits in the context of online dating;" $\alpha = .89$).

A similar approach was taken with a one-way ANOVA with planned contrasts, $F(2, 187) = 46.01$, $p < .001$. Planned contrasts showed that the (novel) inappropriate texts ($M = 3.05$, $SD = 1.52$) scored significantly lower on text appropriateness than the two appropriate text conditions (not novel, appropriate: $M = 5.27$, $SD = 1.06$; novel, appropriate: $M = 4.92$, $SD = 1.52$), $t(187) = 9.47$, $p < .001$, $r = .57$. The ratings of these two appropriate text conditions did not differ significantly on perceived text appropriateness, $t(187) = 1.43$, $p = .16$, $r = .10$.

Dating profiles

All profile texts were accompanied with a blurred profile picture. We decided to include blurred pictures in order to minimize the potential impact of picture attractiveness on the impressions formed of profile owners. The blurred pictures did not reveal the profile owner's gender. Figure 1 shows an example profile in both Dutch and English (translated).

Measures

All nine main variables of this study were measured on a 7-point Likert scale (1 = strongly disagree, 7 = strongly agree).

Expectancy violations

The extent to which the presented profile text aligned/violated online daters' expectations about profile texts was measured with three items,

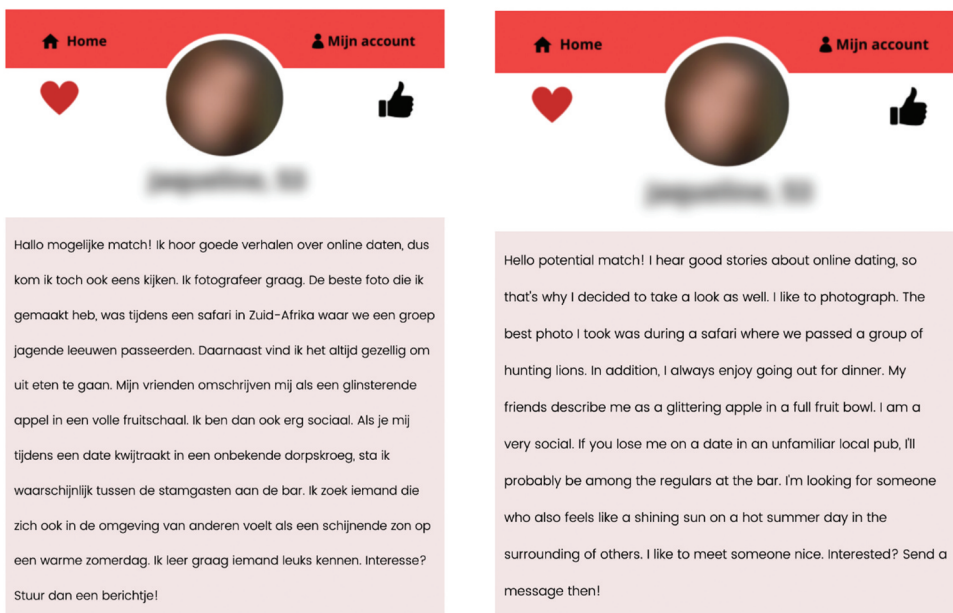


Figure 1. Example of a translated English version (right) of a Dutch novel and appropriate dating profile (left) used in the experiment.

inspired by the violation expectedness items invented by Afifi and Metts (1998). Our three items were “This profile text was as I expected” (reverse-coded), “This profile text surprised me”, and “This profile text did not match my expectations”. Although the Cronbach’s alpha of the scale was only .62, a factor analysis showed that all three items loaded on a single factor, explaining 57.18% of variance with all factor loadings of .71 or higher. The three items were therefore considered as one factor measuring expectancy violations, with higher scores indicating that expectancies were violated more strongly.

Violation valence

Violation valence was measured with three items ($\alpha = .89$) based on Afifi and Metts (1998) and measured the extent to which expectancies were violated positively or negatively (i.e., better/worse than expected). The three items were: “This profile text was disappointing” (reverse-coded), “This profile text surprised me in a good way,” and “This profile text was better than I expected.” The higher the scores, the better the profile text was than expected (i.e., positive expectancy violation).

Perceived profile owner originality

Perceived profile owner originality was measured with three items ($\alpha = .71$). The items were based on the creativity aspect of the Openness to Experience-dimension of the HEXACO from Lee and Ashton (2004) and the Originality facet of Peterson and Seligman’s (2004) Values in Action (VIA) scale, and were “The profile owner seems to be different than most others,” “The profile owner doesn’t seem to be unique” (reverse-coded), and “I think the profile owner has original ideas.”

Perceived profile owner oddness

This study used three items to measure perceived profile owner oddness based on the unconventionality facet of the Openness to Experience aspect of the HEXACO scale (Lee & Ashton, 2004; Simms et al., 2011). The items were “This profile owner seems to be strange/peculiar/normal (reverse-coded)” ($\alpha = .86$).

Perceived romantic attractiveness of profile owner

The perceived romantic attractiveness of the profile owner was measured with three items: “I think the profile owner is attractive,” “I think I could fall for the owner of this profile,” and “I don’t think I could start a relationship with the profile owner” (reverse-coded). The internal consistency of these three items together was fairly low, with a Cronbach’s alpha of .58, which may have been caused by the negative phrasing of the third item (Kamoen et al., 2013). As soon as we removed that item from the scale, a Pearson’s correlation test shows a strong correlation between the two remaining items ($r = .53$, $\alpha = .86$).

We therefore decided to do the analyses clustering these two items, that together represent the level of romantic attraction.

For the measures of perceived profile owner originality, oddness, and romantic attractiveness higher values indicate the profile owner was evaluated as more original, odd, and romantically attractive, respectively.

Procedure

The experiment was created as an online survey in Qualtrics. First, participants were informed about the study's subject, ethics and data management, and were asked for informed consent. Each participant was then randomly allocated to one of the three text conditions ($n_{\text{not novel, appropriate}} = 98$; $n_{\text{novel, inappropriate}} = 95$; $n_{\text{novel, appropriate}} = 102$) and was presented to one of the four profiles from that condition. Next, participants answered three manipulation check items about perceived text novelty, appropriateness, and creativity (measured with one item each), followed by the mediating and dependent variables. Finally, participants were debriefed and received full disclosure. Several other items that are not part of this paper were measured in the survey; these results are available at OSF. The online experiment took around 5–10 minutes to complete.

Data Analysis

Two serial-multiple mediation models were run to test H1abc and H2abc in SPSS using Hayes (2013; model 6) PROCESS macro version 3.1 using 10,000 bootstrap samples and bias-corrected and accelerated (BCa) confidence intervals. The first serial-multiple mediation model tested H1 by examining the effects of novel vs. not novel texts (X) on romantic attractiveness (Y) via expectancy violations (M1) and perceived profile owner originality (M2). The second serial-multiple mediation model tested H2 and looked more specifically at the effects of (in)appropriateness in novel texts (X) on romantic attractiveness (Y) via violation valence (M1) and perceived profile owner oddness (M2).

Results

Text Perceptions and Manipulation Checks

The results of a MANOVA with text condition as independent variable and text novelty, text appropriateness, and text creativity as dependent variables showed that text novelty scores were higher for the novel texts ($M = 4.39$, $SD = 1.72$) than for the not novel texts ($M = 5.63$, $SD = 1.47$), $SE = 0.189$, $p < .001$, $CI [-1.63, -0.88]$. However, different than the pretest results, text novelty scores were also significantly higher for the novel, inappropriate texts ($M = 6.01$, $SD = 1.32$) than

Table 2. Means (SDs) on the three text conditions for all included variables.

Type of variable	Variable	Not novel, appropriate (<i>n</i> = 98)	Novel, inappropriate (<i>n</i> = 95)	Novel, appropriate (<i>n</i> = 102)
Text perceptions	Novelty	4.39 (1.72) ^a	6.01 (1.32) ^b	5.27 (1.52) ^c
	Appropriateness	5.35 (1.47) ^a	3.27 (1.78) ^b	5.06 (1.63) ^a
	Creativity	3.33 (1.61) ^a	3.71 (1.92) ^a	4.58 (1.58) ^b
Expectancy violations	Violations	3.79 (1.14) ^a	5.31 (1.16) ^b	4.64 (1.08) ^b
	Valence	3.52 (1.63) ^a	2.72 (1.49) ^b	3.76 (1.55) ^a
Person perceptions	Originality	3.43 (1.27) ^a	4.51 (1.23) ^b	4.15 (1.36) ^c
	Oddness	2.88 (1.09) ^a	4.92 (1.36) ^b	3.76 (1.40) ^c
	Creativity	3.53 (1.50) ^a	3.92 (1.46) ^a	4.34 (1.53) ^b
	Romantic attractiveness	3.48 (1.23) ^a	2.91 (1.35) ^b	3.43 (1.28) ^a

Note. Different superscripts indicate significant differences between the conditions in that row. All variables were measured on a scale from 1 to 7.

for the novel, appropriate texts ($M = 5.27$, $SD = 1.52$), $SE = 0.218$, $p < .001$, $CI [0.31, 1.17]$.

Moreover, as found in the pretest, novel texts that were inappropriate scored lower on perceived text appropriateness ($M = 3.27$, $SD = 1.78$) than those that were appropriate ($M = 5.06$, $SD = 1.63$), $SE = 0.232$, $p < .001$, $CI [-2.24, -1.33]$. Texts that were not novel and appropriate were considered as even more appropriate ($M = 5.35$, $SD = 1.47$) than the novel ones ($M = 4.20$, $SD = 1.92$), $SE = 0.201$, $p < .001$, $CI [0.79, 1.58]$.

Results on text creativity perceptions show that novel texts ($M = 4.16$; $SD = 1.80$) score higher on text creativity than those that are not novel ($M = 3.33$; $SD = 1.61$), $SE = 0.211$, $p < .001$, $CI [-1.23, -0.40]$. Furthermore, profiles that meet both criteria of creativity, and are thus novel and appropriate, score significantly higher on perceived text creativity ($M = 4.58$, $SD = 1.58$) than texts that are novel and inappropriate ($M = 3.71$, $SD = 1.92$), $SE = 0.243$, $p < .001$, $CI [-1.35, -0.39]$.

The mean scores and standard deviations for all three text conditions for all manipulation check, mediating and outcome variables can be found in Table 2, and the in Table 3.

Table 3. Pearson correlation coefficient values for all study variables.

Variable	1	2	3	4	5	6	7
1. Text novelty							
2. Text appropriateness	-.35**						
3. Text creativity	.23**	.22**					
4. Expectancy violations	.57**	-.41**	.15**				
5. Violation valence	.07	.42**	.65**	-.11			
6. Person originality	.44**	-.07	.50**	.43**	.42**		
7. Person oddness	.37**	-.61**	-.16**	.53**	-.47**	.17**	
8. Romantic attractiveness	-.12*	.41**	.37**	-.18**	.60**	.25**	-.44**

**correlation is significant at the .01 level, * correlation is significant at the .05 level.

Table 4. All indirect effects of novelty (Nov.) and appropriateness (Appr.) on romantic attractiveness (R_Attr.) through expectancy violations (EV) and profile owner originality (orig.) for novelty and through violation valence (VV) and profile owner oddness (odd) for appropriateness.

Contrast	Effects	Product of coefficients		Bootstrapping 95% BCa Confidence Interval	
		β	SE	Lower	Upper
Not novel (<i>n</i> = 98) vs. novel (<i>n</i> = 197)	Total indirect effects	−0.02	0.08	−0.18	0.15
	Nov. >> EV >> R_Attr.	−0.29	0.07	−0.45	−0.16
	Nov. >> Orig. >> R_Attr.	0.13	0.05	0.03	0.25
	Nov. >> EV >> Orig. >> R_Attr.	0.14	0.04	0.08	0.22
Novel, inappropriate (<i>n</i> = 102) vs. novel, appropriate (<i>n</i> = 95)	Total indirect effects	0.49	0.10	0.31	0.68
	Appr. >> VV >> R_Attr.	0.32	0.08	0.18	0.49
	Appr. >> Odd >> R_Attr.	0.10	0.05	0.02	0.20
	Appr. >> VV >> Odd >> R_Attr.	0.08	0.03	0.02	0.15

Testing H1abc

The first model tested the first set of hypotheses on novel vs. not novel texts, and more specifically tested the extent to which novel texts positively affect romantic attractiveness through increased scores on expectancy violations (M1) and perceived originality (M2). All results and indirect effects of the serial mediation models are presented in Table 4.

In line with H1a, results indeed show that novel texts ($M = 4.96$, $SD = 1.17$) score higher on expectancy violations than not novel texts ($M = 3.79$, $SD = 1.14$), $\beta = 0.91$, $SE = .14$, $p < .001$, indicating that novel texts violate expectancies more than not novel texts. When texts violated expectancies more, their writers score higher on originality, $\beta = 0.37$, $SE = .06$, $p < .001$, which aligns with H1b. Moreover, profile owners who score higher on originality are also perceived as more romantically attractive ($\beta = 0.42$, $SE = .06$, $p < .001$), which supports H1c.

All indirect effects were significant, but not all in the same direction. There was a negative effect of text novelty on romantic attractiveness via expectancy violations, $\beta = -0.29$, $SE = 0.07$, CI $[-0.45, -0.16]$. Novel texts violate expectations more than non-novel texts, resulting in lower romantic attractiveness. However, there was also a positive indirect effect of novelty via expectancy violations and perceived originality on romantic attractiveness ($\beta = 0.14$, $SE = 0.04$, CI $[0.08, 0.22]$), as well as a positive effect of novelty directly on perceived originality and romantic attractiveness, $\beta = 0.13$, $SE = 0.05$, CI $[0.03, 0.25]$. Thus, novel texts created a perception of an original profile owner, either directly or indirectly via expectancy violations, which in turn increased the profile owner's romantic attractiveness. Figure 2 demonstrates the results of the serial-multiple mediation for text novelty.

When we compare only the novel, appropriate text condition with the non-novel text condition in an additional mediation analysis, we find similar results: a direct negative relationship between text novelty and romantic attractiveness and a positive double mediation via expectancy violations and

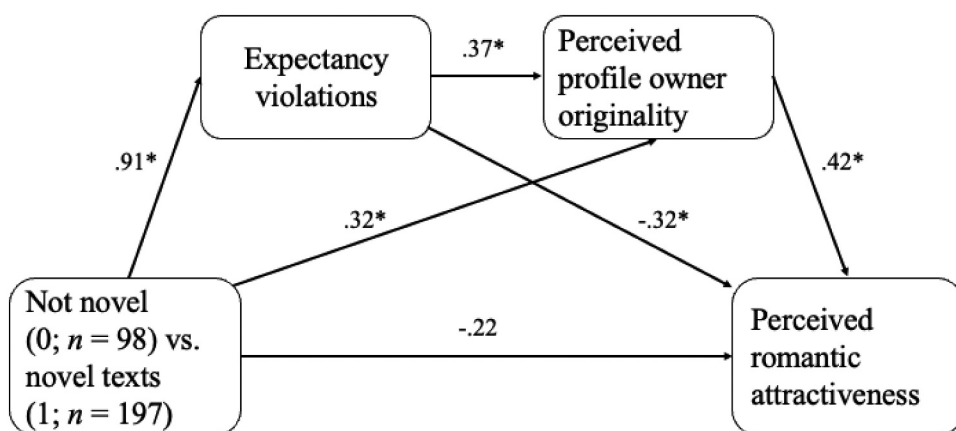


Figure 2. Serial-multiple mediation of expectancy violations and perceived originality in the relationship between text novelty and perceived romantic attractiveness with standardized coefficients. *indicates $p < .05$.

profile owner originality. However, the significant mediated relationship between novelty and romantic attractiveness via originality alone is no longer significant.

Testing H2abc

The second serial-multiple mediation model compared the novel texts that were inappropriate with those that were appropriate, and examined the extent to which novel, inappropriate texts affect perceived romantic attractiveness through decreased scores on violation valence (M1) and increased perceived oddness (M2). As stated in H2a, novel and inappropriate texts ($M = 2.72$, $SD = 1.94$) indeed affected expectancy violations more negatively than the novel and appropriate texts ($M = 3.76$, $SD = 1.55$), $\beta = 0.65$, $SE = .22$, $p < .001$. Violation valence was in turn negatively related to profile owner oddness, $\beta = -0.52$, $SE = .06$, $p < .001$, confirming H2b. In turn, perceived oddness related negatively to perceived romantic attractiveness, $\beta = -0.22$, $SE = .06$, $p = .002$, supporting H2c.

All indirect effects were significant. Appropriate texts resulted in more positive violation valence than inappropriate texts, which resulted in higher perceived romantic attractiveness of the profile owner, directly ($\beta = 0.32$, $SE = 0.08$, $CI [0.18, 0.49]$) as well as through reduced perceptions of oddness ($\beta = 0.08$, $SE = 0.03$, $CI [0.02, 0.15]$). Appropriate texts also directly decreased oddness perceptions, increasing romantic attractiveness accordingly ($\beta = 0.10$, $SE = 0.05$, $CI [0.02, 0.20]$). Figure 3 presents the results of the serial-multiple mediation for text appropriateness.

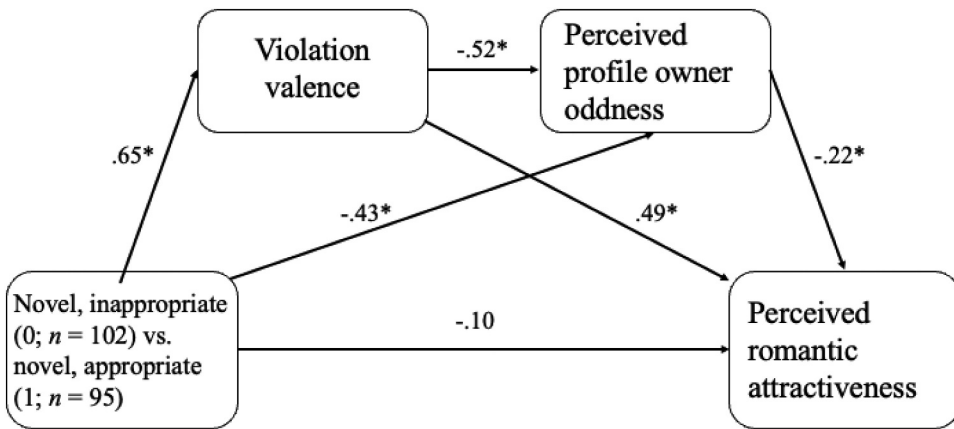


Figure 3. Serial-multiple mediation of violation valence and perceived oddness in the relationship between text appropriateness and perceived romantic attractiveness with standardized coefficients. *indicates $p < .05$. *Note.* The positive β from condition to violation valence demonstrates that novel, appropriate texts violated expectancies more positively than novel, inappropriate texts. In other words, inappropriate texts were worse than expected and violated expectancies negatively.

Discussion

This study investigated how novelty and appropriateness, the two dimensions of creativity, influence perceptions of the romantic attractiveness of owners of online dating profile texts, and to what extent expectancy violations and perceptions of the profile owner's originality and oddness explain these impressions of romantic attractiveness. We expected that novel profile texts would violate expectations more than not novel texts, but that the valence of this violation would depend on the text's appropriateness. Specifically, novel texts would violate expectations of the common profile texts, causing the profile owner to be seen as more original and hence more romantically attractive. Compared to novel but appropriate texts, though, inappropriate texts would violate expectations negatively, resulting in the profile owner to be judged as odd, thereby decreasing romantic attractiveness. Actual online daters were therefore presented with a dating profile that met either one or two of the criteria of creativity and rated the text and profile owner afterward.

Our expectations were generally confirmed. Novel texts violated expectations. These violated expectations increased perceptions of a profile owner's originality, which in turn increased perceptions of profile owners' romantic attractiveness. Appropriate texts resulted in more positive violation valence than inappropriate texts. Positive expectancy violations resulted in increased romantic attractiveness, both directly and through reduced perceptions of oddness. These results indicate that novelty and appropriateness in an online dating profile serve as cues to form impressions about the texts and their writers.

Regarding novelty, our results were both confirmed and not confirmed. In line with our hypotheses, we found that novel texts violated expectations more than not novel texts. Novel profile texts indeed seem to violate expectations of what is normal in dating profiles. This violation leads people to perceive the profile owner as more original, which in turn affects romantic attractiveness. That owners of original profiles are perceived as more attractive corresponds with findings of Van der Zanden et al. (2022), who showed that owners of profiles assessed as more original tend to be evaluated more positively on other personality attributes as well, ultimately affecting perceptions of profile owner attractiveness.

However, the results also show a negative indirect effect of novelty (regardless of the appropriateness condition) on romantic attractiveness via expectancy violations. Some users may prefer novel profile texts, as they may value creativity in a partner. These users are positively surprised by the expectancy violation and thus find the writer of this novel text romantically attractive. Other users may less appreciate a novel text, as an (overly) creative profile owner may not match their preferences and personality. This finding corroborates those of DelGreco and Denes (DelGreco & Denes 2020) who found that women who responded to a compliment conform expectations (i.e., accepting by just thanking) were evaluated more likeable and socially attractive than those violating expectations positively or negatively. This together suggests that some people prefer the mundane and specifically look for dating partners who not deviate too much from the norm.

Regarding appropriateness, our study shows that appropriateness explains the positive and negative effects of text novelty on romantic attractiveness. That is, owners of novel but inappropriate profiles were seen as less romantically attractive than owners of profiles that are novel *and* appropriate. Specifically, inappropriate texts led to more negative expectancy violations than appropriate texts. Similarly, appropriate texts lead to more positive expectancy violations than inappropriate texts. The more positive the expectancy violation, the less profile owners are considered odd, and the more they are judged as romantically attractive. These results thus confirm our hypotheses of text appropriateness and earlier findings (Van der Zanden et al., 2022; White et al., 2018). Moreover, inappropriate texts also directly led profile owners to be evaluated as odder. This shows that inappropriate profile texts are not only perceived as odd because they violate expectations of what is normal on a dating platform, but probably also because people feel this text reflects the profile owner's personality. Moreover, a positive violation valence also directly affected perceived romantic attractiveness (and not only via perceived oddness) as oddness is most likely not the only personality factor that is affected by the text's appropriateness. For example, owners of appropriate profile texts may also be considered more polite and/or caring, which may also positively affect attractiveness.

Implications, Limitations, and Future Research Directions

This study yields several theoretical implications. First, our findings contribute to existing creativity research that extensively investigated novelty and appropriateness as dimensions of creativity, but often neglected the potential effects of the two on impressions formed of the producers of creative products, such as dating profile texts. This study shows that creativity is a balancing act between novelty and appropriateness, at least in the context of online dating profiles. Novelty is the necessary ingredient of creativity. Compared to writers of mundane texts, those of novel texts are perceived as more original. This mirrors previous studies that identified novelty as the first-order criterion of creativity (e.g., Diedrich et al., 2015). Appropriateness, though, reflects how creativity is ultimately judged. Inappropriate dating profile texts are deemed strange and lower perceptions of romantic attractiveness. Therefore, for creativity to work, both novelty and appropriateness are paramount. In the context of online dating, this means that novelty may be enough to grab attention, but appropriateness may determine how a novel text is ultimately valued. Still, novelty and appropriateness cannot be fully separated as distinct dimensions of creativity. The results on text perceptions in the main experiment show that both dimensions determine (to some extent) perceptions on the other dimension: appropriateness influences text novelty perceptions, and novelty impacts text appropriateness perceptions.

Second, the positive and negative effects of novelty on romantic attractiveness via expectancy violations and perceived originality indicate that there may be different groups of online daters who perceived novelty in dating profile texts differently. For some people a novel text is mainly seen as an expectancy violation, lowering romantic attractiveness. For others, the violation expectancy is also noticed, but seen as original, boosting romantic attractiveness. Some people thus appreciate originality while others may appreciate familiarity. These two routes from novelty to romantic attractiveness ask for a further investigation on which characteristics determine the group that appreciates novelty, and the group that dislikes novelty but may appreciate other dating text characteristics. For example, people who are more risk-averse may be less likely to seek out others with creative texts, as they are not looking for someone outside their comfort zone.

Third, we show that if novel texts do not conform to social norms and expectations and violate these expectations negatively, this harms the profile owner as they are consequently perceived as odder and thus as less romantically attractive. This implies that appropriateness is highly important when it comes to perceptions of romantic attractiveness. While appropriateness may play a secondary role in determining creativity, it seems to play a primary role in determining romantic attractiveness: if only one of the two criteria can be fulfilled, it might be more rewarding to be

appropriate (and not novel) than to be novel (and inappropriate). An interesting next step would be to further examine how social norms and expectations differ from platform to platform, and how the similarities and differences in the expectations on these platform types affect impression formation accordingly. That is, our findings are based on a sample of middle-aged adults who are active on a more traditional Dutch dating site and can thus not be generalized to all online daters. Their expectations may differ from those on dating applications, for example because the textual component plays a smaller part on such apps and/or the demographic and basic characteristics of the users are different. What our group of participants may deem novel and appropriate may not be similar to what, for example, younger online daters deem novel and appropriate. Also, the general norm on an online dating platform may affect what is appropriate or not; dating platforms that are considered hookup platforms (e.g., Ashley Madison) may have different norms of appropriateness than platforms that focus on committed relationships.

As a further limitation to our study, it should be mentioned that the internal consistency of some of our scales was relatively low (i.e., romantic attraction and expectancy violations), though the romantic attractiveness scale was adequate when we removed the negatively worded item. Moreover, the scale we used for expectancy violations was based on a validated scale (Afifi & Metts, 1998), although we did not include all items in our scale and reformulated some items to fit our study. Possibly, future research could use different operationalizations of these constructs in order to replicate and extend our results.

Our research affirms the relevance of the expectancy violations theory to explain impression formation in online dating. Moreover, our findings demonstrate that expectations already play a role in the initial phase of profile evaluation – prior to interaction. This focus diverges from the limited body of prior online dating research using EVT that focused on expectancies in the online interaction phase (DelGreco & Denes, 2022) or the switch from online to offline interaction (Ramirez & Wang, 2008; Zhao & Yan, 2022). These studies were thereby concerned with the expectations online daters hold about specific individuals (Ramirez & Wang, 2008; Zhao & Yan, 2022) or specific groups (e.g., women; DelGreco & Denes, 2022) in interpersonal communication, but not, as in our study, with broad expectations based on what is generally encountered, and thus expected, in profile texts. This prompts many interesting future research avenues, potentially examining general expectations of users' profiles picture or delving into the match between pre-interaction expectations (when viewing someone's profile) and online interaction outcomes (when exchanging messages).

Furthermore, our findings also extend EVT more generally. First, we have shown that EVT may be used to study creativity by exploring its link with (the

dimensions of) creativity and impression formation. For example, EVT may be used to study creativity perceptions of other user-generated messages, such as social media content or job application letters, but also in wider communicative messages, such as in literature, art, and music. Evaluations of these artistic expressions may also be a function of expectancy violations and violation valence. Second, we demonstrate that EVT can also be applied outside of the interpersonal communication domain. That is, our study shows that the framework can also be used to study (online) impression formation, before interaction has taken place, based on passive or third-party cues.

In addition to theoretical implications, this study also reveals practical insights that can help online daters who aim to create a creative profile text. Based on our findings, they are advised to construct profiles that are novel – meaning different from most other profiles – as these are unexpected, and this enhances perceptions of originality and perceived romantic attractiveness. Novelty can be, for instance, be brought about by including metaphorical expressions in a profile (e.g., “I am looking for someone as warm as a crackling fireplace” instead of “I am looking for someone who is friendly and kind”) and by including more concrete self-disclosure statements (e.g., “Three times a week I help my mom with grocery shopping” instead of “I am a caring person”). However, if one decides to construct a novel text, the social norms and expectations that exist within the online dating context should always be kept in mind, since not adhering to them will violate expectations negatively, and make you appear more odd and less romantically attractive. More specifically, appropriateness can be constituted by using positively framed and not too intimate self-disclosure (e.g., not “Looking for someone who can make me a good lasagna with bechamel sauce” and “I am taking care of my mom who has cancer since 2019”).

It is important to realize though, that creativity and its effects are context-dependent (Corazza, 2016). What is novel and appropriate varies among different dating platforms, across cultures and groups, and over time. If many online daters would read this paper and change their profile texts accordingly, the collective perception of what constitutes a novel dating text would change, and novelty would no longer be defined as we did in this paper. In this study, we manipulated novelty and appropriateness in the profiles in two ways: through metaphors – a stylistic characteristic mostly related to form – and through concrete self-disclosure – a characteristic mostly related to meaning. While these manipulations seemed to be successful, the distinct impacts of metaphors and (types of) concrete self-disclosure remain yet unknown. That is because to manipulate text novelty and appropriateness we took into account both elements simultaneously. An interesting suggestion for future research would be to explore if one of the two elements impacts impression formation and attractiveness perceptions more strongly. Employing

interviews or thinking-aloud methods would be one useful way to get a more comprehensive picture of how online daters pick up and use these cues during impression formation processes and what specifically underlies the impressions people form of texts that are (not) novel and (in) appropriate.

Conclusion

The present study is one of the first that examined whether and how novelty and appropriateness – the two criteria of creativity – violate expectancies and how the two dimensions are used as cues to form impressions of the personality and, consequently, the romantic attractiveness of dating profile owners. Our results suggest that novel texts are less expected, and that this enhances perceived profile owner originality, which, in turn, enhances perceived romantic attractiveness. Moreover, novel texts that are inappropriate negatively impact impression formation as these profiles violate expectations more negatively, resulting in higher oddness and lower perceptions of romantic attractiveness. With our findings, we (1) extend expectancy violation theory by showing that the theory can also be used to explain initial impression formation based on passive cues instead of only being applicable in interpersonal communication, (2) demonstrate that EVT may be used to study (text) creativity, (3) show that creativity consists of two distinct but related dimensions, and (4) show that creativity can be an important concept for further developing theories of interpersonal attraction and impression formation.

Notes

1. In our preregistration, we formulated an additional hypothesis (H5) concerning the relationship between novelty and four different personality perceptions. However, due to word limitations, this is not discussed in the current manuscript. Findings regarding this hypothesis can be found on OSF.
2. The numbering and ordering of the hypotheses in this manuscript differs from the preregistered order to retain the manuscript's flow.

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Disclosure Statement

No potential conflict of interest was reported by the authors.

Data Availability Statement

The data that support the findings of this study are available on the Open Science Framework via: <https://osf.io/sympn/>

Open Scholarship



This article has earned the Center for Open Science badges for Open Data, Open Materials and Preregistered. The data and materials are openly accessible at <https://osf.io/sympn/>.

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